

Macroeconomic policies

Fiscal policy and the budget

Fiscal policy is the use of taxation and government spending to manage aggregate demand in order to achieve the government's macroeconomic aims. The government's annual budget is a statement of its fiscal policy. The budget often receives much media attention as it is an indicator of both fiscal policy intentions and economic performance. In the budget statement, the finance minister outlines the government's spending and taxation plans for the year ahead. A budget surplus arises when tax revenue exceeds government spending. In contrast, a budget deficit occurs when government spending exceeds tax revenue and a balanced budget is when government spending matches tax revenue.

Most governments seek to achieve a balanced budget over time. In the short term, a government may aim for, or welcome, a budget deficit if there is a low level of economic activity. A budget deficit may occur in this situation as a result of both deliberate government action and of automatic stabilizers. If there is a decline in economic growth and a rise in unemployment, a government may decide to cut tax rates and increase government spending. It may also allow government spending on unemployment benefits to rise and tax revenue to fall as an automatic result of a slowdown in the economy. A budget deficit that occurs due to a fall in economic activity is known as a cyclical deficit. A government is unlikely to be concerned about a cyclical deficit as it will move towards a balance as economic activity increases.

However, a government will be concerned about a structural deficit. A structural deficit arises when a government is committed to too much spending relative to its tax revenue. In this case, the deficit will not disappear when GDP increases. In practice, a budget deficit may contain both cyclical and structural elements.

A budget deficit will decline, at least in the short run, if there is a rise in tax revenue and/or a fall in government expenditure. However, there is the possibility that a rise in government spending and/or a cut in tax rates or the tax base could reduce a budget deficit. This is because such a change/changes could increase economic activity. If, for instance, increased government expenditure on training made workers more skilled, they may earn higher wages and so pay more taxes.

The national debt

The national debt is often expressed as a percentage of GDP. Government or public sector debt is the total debt a central government, or the whole public sector, has built up over time. It is connected to budget deficits and budget surpluses. If a government has a budget deficit in one year it will add to the country's national debt. In contrast, the extra revenue earned from a budget surplus can be used to pay off part of the national debt.

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The national debt tends to increase during economic downturns as this is when government expenditure tends to rise at a more rapid rate than tax revenue. There may also be a tendency for a government to spend more than its revenue even during economic booms (a structural deficit). In addition, military conflicts can result in significant increases to the national debt.

There are disadvantages in having a large and increasing national debt. There is the opportunity cost of interest payments on the national debt. The government revenue used to service the debt might have been used to finance, for instance, the building of new hospitals. A large national debt may make financial institutions, firms, individuals and foreign governments reluctant to lend. This is because it may cause them to have doubts about the government's ability to pay interest on the debt and to repay the sum borrowed. Borrowing a large amount may also push up the rate of interest that has to be paid.

National debt is not the same as external debt as some of the debt will be owed to citizens of the country. For example, some households may have purchased government saving certificates and some domestic banks may have bought government bonds. However, payments to foreign lenders by the government will involve an outflow of money from the country.

The national debt to GDP ratio will be reduced if there is either some repayment of the debt or if GDP rises.

Sin taxes

Indirect taxes on particular products are sometimes known as excise duties. Some excise duties are sometimes referred to as sin taxes. Sin taxes are imposed to discourage people from buying products that are not good for their health. For example, a number of countries, including Denmark, France and Mexico, have imposed taxes on high-fat foods.

Key terms:

Sin taxes: taxes on products considered harmful to consumers

Indirect taxes versus direct taxes

In recent decades there has been a tendency for governments to rely more on indirect taxes. Indirect taxes have a number of advantages. They can be changed, if needed, relatively quickly and easily. They are cheaper to collect than direct taxes as firms do part of the administrative work. They can also be used to discourage the purchase of particular products.

The main advantage claimed for indirect taxes is that they do not discourage effort, innovation and saving. Higher direct taxes may put off some people from joining the labour

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force, may stop some people working overtime and may encourage some people to cut the standard hours they work. This is because the disposable income people earn for each hour they work will be reduced.

Direct taxes may act as a disincentive to save as they mean that income can be taxed twice – once when it is earned and again when interest is received on any part that is saved. High direct taxes may stop firms introducing new methods and products if they think their post-tax income will be too low.

However, a rise in direct taxes will not necessarily have a disincentive effect. Some workers may decide to work more hours to maintain their level of disposable income. High direct taxes may encourage tax avoidance and tax evasion.

Greater reliance on indirect taxes may make income less evenly distributed. This is because indirect taxes are regressive taxes. Paying a tax of, for example, \$2 on a product will take up a larger percentage of the income of a person on a low income than that of a rich person. Indirect taxes can also be inflationary, pushing up the price level.

The imposition of or increase in indirect taxes can contribute to cost-push inflation. Indirect taxes place an extra cost on suppliers. They are likely to result in higher prices and this may lead people to expect prices to continue to rise. High rates of indirect taxes may also encourage people to try to avoid paying them. For instance, people may smuggle products into the country to avoid paying import tariffs.

Progressive, regressive and proportional taxes

While indirect taxes are regressive, direct taxes are usually progressive. A progressive tax is one that takes a higher percentage of a person or firm's income as that income rises. Table 22.3 gives an example of a progressive tax.

Income range (\$)	Tax rate (%)	Tax paid (\$)
0 – 8 000	0	0
8 001 – 20 000	25	3 000
20 001 – 50 000	40	11 000
50 001 +	50	

Table 22.3: Example of a progressive income tax

In this system, people earning \$20 000 pay \$3000 in tax (25% of \$12 000). This is 15% of their income. People earning \$40 000 pay \$11 000 (\$3000 + 40% of \$20 000 = \$8000). This is 27.5% of their income. Those who earn \$80 000 pay 32.5% of their income (\$3000

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+ \$8000 + \$15 000 (50% of \$30 000) = \$26 000). So as income rises, a higher percentage is paid in tax.

In contrast, in the case of a regressive tax, a smaller percentage of income is taken as income rises. This means that such a tax takes a higher percentage of the income of people on low incomes.

A proportional tax is a fixed percentage tax, for example a 20% income tax. The tax rate does not change as income changes. In recent years, a number of countries, including Estonia and Russia, have introduced a flat rate tax system. A 'pure' version of this system would be one tax rate, for example an income tax rate of 15%, a corporate tax rate of 15% and a sales tax rate of 15%. Such a tax system is simple to understand and administer, but is regressive.

Marginal and average rates of taxation

The marginal rate of taxation (mrt) is the proportion of extra income taken in tax. For example, if a person earns an extra £100 and £30 is taken in tax, the marginal tax rate is $\text{£}30/\text{£}100 = 0.3$. This can also be expressed as 30%. In contrast, the average rate of taxation (art) is the proportion of a person's total income that is taken in tax. If a person earns £50 000 and pays £10 000 in tax, the average tax rate is 0.2 or 20%.

In the case of a progressive tax, the marginal tax rate is higher than the average tax rate. The proportion of tax people would pay on extra income would be greater than the proportion they pay on the total amount they earn. In the example, given above on a progressive tax, people earning \$80 000 have a marginal tax rate of 50% but an average tax rate of 32.5%.

In contrast, in the case of a regressive tax, the marginal tax rate is lower than the average tax rate. The relationship is also different in the case of a proportional tax. In this case, the marginal tax rate equals the average tax rate.

The reasons for taxation

Taxes are imposed for a number of reasons. One is to raise revenue to finance government spending on merit goods, such as education, and public goods, such as defence. The government also imposes taxes to influence aggregate demand. If the government wants to reduce aggregate demand it will raise tax rates and/or increase the tax base. If it wants to increase aggregate demand, it will do the reverse. The need to influence aggregate demand is probably the main reason for imposing taxes. A government could, in theory, finance its spending just by printing money. However, such an approach would be very inflationary. The government needs to reduce public sector demand to free up resources for use by the government.

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A progressive income tax may be used to distribute income more evenly. Such a tax narrows the gap between the disposable income of rich people and people on low incomes. The gap could be further narrowed by the government using some of the tax taken from rich people to provide cash benefits to those on low incomes.

Taxes are imposed to discourage the consumption of certain products. These may be imports. A government may be concerned that the country is spending more on imports than it is earning from selling exports. Taxes may also be imposed on demerit goods in order to improve people's health and the environment.

Government spending

Government spending can be divided into spending on transfer payments (not counted in measures of national income), current spending and capital spending, as shown in Figure 22.7.

Government spending on transfer payments (welfare payments to certain groups of people) includes spending on unemployment benefits, state pensions and interest payments on the national debt.

Current government spending is spending on goods and services used to provide state-financed services. Current government spending covers the operating costs, for instance, the spending on the wages of teachers employed in state schools and the medicines used in state hospitals.

In contrast, capital government spending is spending on capital goods used in the public sector. Capital government spending includes, for instance, spending on building state schools and hospitals.

Government spending can also be divided into exhaustive and non-exhaustive spending. Exhaustive government spending covers current and capital spending. It is spending which uses resources and is counted in aggregate demand and GDP. Non-exhaustive government spending is spending on transfer payments. This spending does not involve the government deciding how resources are used. The people who receive the payments make the decision about how to use the resources.

Reasons for government spending

The reasons for government spending are linked to the reasons for taxation. Governments spend in order to influence aggregate demand and so the level of economic activity. If private sector spending is thought to be too low, a government may decide to inject more spending into the economy. It will aim for a budget deficit, spending more than it raises in taxation.

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As well as seeking to influence aggregate demand, a government may use its spending to increase aggregate supply. Government spending on education, healthcare and infrastructure can raise an economy's productive potential.

Spending on transfer payments is designed to ensure that people have a basic level of income, and so avoid poverty, and to reduce income inequality. Unemployment benefits, for instance, provide an income for those who have lost their jobs and run the risk of falling into poverty.

Governments also spend on merit and public goods in order to overcome market failure. As GDP rises, there is likely to be pressure on the government to spend more on merit and public goods. This is because people will have higher expectations of, for example, the quality of education they receive. There are also likely to be advances in technology which make it possible to undertake more complex operations that also require more expensive aftercare that will increase spending on healthcare.

In practice, governments may also spend to win political popularity so that they can stay in power. Government spending in a number of countries regularly rises just before an election. Governments may also be influenced by pressure groups that encourage them to spend more on, for instance, the environment.

The tools of monetary policy

The main tools of monetary policy are:

- **Interest rates.** In recent years, in a number of countries, changes in interest rates have been the main monetary policy tool that central banks have used to control inflation and to influence economic activity. The interest rate is, in effect, the price of money. Households and firms who want to borrow money have to pay interest. Households and firms who lend money are paid interest. The interest rate charged by the central bank may be called the bank, base, repo or often just the interest rate. Changes in the interest rate have been mainly used to achieve price stability. Since 2008, however, the governments of a number of countries have begun to prioritize the encouragement of economic growth. They include Japan, the USA and the UK.
- **The money supply.** A central bank may also target the money supply in the economy as changes in the quantity of money in the economy can influence aggregate demand. A central bank can electronically print money but the main cause of changes in the money supply is lending by commercial banks. It is for this reason that central banks often seek to influence lending by commercial banks.

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- **Exchange rate.** Most economists also include the exchange rate as a monetary policy tool. Central banks may manipulate the exchange rate to raise or lower aggregate demand and so influence, for example, price stability.
- **Credit regulations.** Central banks may also impose credit regulations on commercial banks to help maintain financial stability and to influence bank lending. Most central banks require the country's commercial banks to hold a proportion of their assets in a form that can be quickly sold and so converted into cash. This is to ensure the commercial banks can meet their customers' likely demand for cash even during a financial crisis.

The effectiveness of monetary policy

In practice, it can be difficult to control the money supply. This is partly because commercial banks have a strong incentive to try to increase their lending and may seek to get round any limits a central bank seeks to place on the growth of their lending. Trying to control certain forms of money may lead to new forms of money being used.

Using interest rates is also not without problems. There is a time lag between changing interest rates and the full effect being transmitted to the macroeconomy. Some economists have estimated that it can take as long as 18 months for interest rate changes to have their full impact. This is, however, less time than in the case of some fiscal policy measures.

Interest rate changes may be a powerful policy measure but they are also a blunt and uncertain one. Some households and firms are more likely than others to cope with a change in the interest rate. For example, a rise in the rate of interest harms borrowers but benefits savers. A higher interest rate may also have an adverse effect on unemployment and economic growth, as with a deflationary fiscal policy measure.

With increasing mobility of financial investment, it can be difficult for a country to operate an interest rate that is significantly different from rival countries.

Supply-side policy tools

The tools of supply-side policy include spending on education and training, promoting infrastructure development and support for technological improvement. Other tools include cuts in corporate tax, cuts in income tax, trade union reform, privatization, deregulation and relaxation of immigration controls.

Education and training

Increasing spending on education and training may raise the quality of education and training. If this occurs, workers' skills and productivity may increase as well as their flexibility and mobility. A more efficient labour force is one of the key causes of an

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increase in a country's productive capacity. With the same sized labour force, more goods and services can be produced. A better educated and trained population can also increase the quality of entrepreneurship, which can increase innovation.

Promoting infrastructure development

Good quality infrastructure, including efficient transport, power, energy and telecommunication networks, keeps the costs of firms low and enables them to get their products to market quickly. For instance, reducing power outages will mean that production is interrupted less frequently. Improving the rail network will reduce the costs of moving products to and from firms and will reduce the chance of workers arriving late.

The government may finance and provide transport infrastructure itself or it may encourage the private sector to provide infrastructure such as a motorway which charges motorists a fee for using it.

Support for technological improvement

Technological improvement can enable capital equipment to produce a greater output at a lower cost. A government can subsidise both universities and private sector firms to encourage the development and introduction of new technology. Government subsidies are used widely throughout the world to encourage firms to increase output.

Other tools

- Cuts in corporate tax. Cutting corporate tax may encourage investment, as firms will have more funds to invest and they will know that they will be able to keep more of any profit earned. More investment will increase both aggregate demand and aggregate supply.
- Cuts in income tax. Cutting income tax may encourage workers to increase their working hours and accept promotion and greater responsibility. It may also persuade some workers to stay in the labour force for longer and persuade others to enter the labour force.
- Trade union reform may increase workers' flexibility and mobility and cut down on the number of days lost through strikes. A resulting fall in industrial action may, in addition, make multinational companies more willing to invest in the country. It is possible that the measure may increase both productivity and production.
- Privatisation and deregulation. In recent years, many countries have adopted privatisation programmes in the belief that firms operate more efficiently in the private sector. Some have also deregulated a number of markets by removing barriers to entry and laws and regulations that increase firms' costs of production.

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- Encouragement of immigration. If a government encourages immigration of skilled workers, it can increase both the quantity and quality of labour in the country.

The impact of supply-side policy tools on the macroeconomy

Supply-side policy tools have the potential to benefit all of a government's policy objectives in the long term. They can raise economic growth, reduce inflationary pressure and lower unemployment.

The impact on national income and real output

By increasing the productivity of labour and capital resources, supply-side policy tools can increase a country's national income and real output. They can do this by increasing the quality and quantity of resources. For example, improved training should enable workers to produce more goods and services. Government spending on infrastructure will increase the quantity of resources that can be used in the production and transport of products.

Technological improvement raises the quality and productivity of capital goods. For example, robots that pick fruit are being introduced into farming. These are increasing output and reducing the costs of production.

The impact on the price level

Over time, aggregate demand tends to increase. If increases in aggregate supply can keep pace with higher aggregate demand, a country can enjoy higher output (higher real GDP) without experiencing demand-pull inflation. This situation is shown in Figure 24.3.

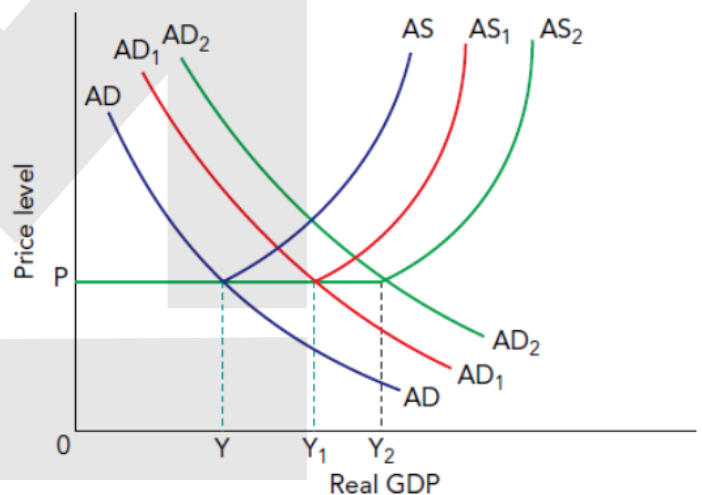


Figure 24.3: Non-inflationary economic growth

Governments may also employ supply-side policy measures to correct cost-push inflation. For example, increased spending on training can raise labour productivity and so reduce labour costs or at least reduce the upward pressure on labour costs.

The impact on employment

Supply-side policy tools can influence employment in a number of ways. Improved education and training will increase workers' skills and geographical and occupational mobility. This should reduce both frictional and structural unemployment. Technological improvement may result in some workers being replaced by capital equipment. However, it is possible that both technological improvement and infrastructure development may increase employment. This is because both lower costs of production, which may increase sales of

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goods and services at home and abroad. More workers may be taken on to produce the higher output.

The effectiveness of supply-side policies

Spending on education and training has the potential to be very effective in the long run. This is because such spending may directly increase the quality of labour and productive capacity. However, it may not be very effective in the short run, as it can take a long time to have an effect. In the short run, it may also contribute to inflation. This is because increased government spending on education and training may increase aggregate demand before it increases aggregate supply.

Providing more education and training may not be very effective if it is not of a high quality or it develops skills that will not be in demand in the longer term. In addition, increased spending on training may be successful in raising the skills of workers, but if their pay rises by more than their productivity, costs of production will still rise.

Infrastructure development can be expensive and can take considerable time to construct. For example, it may take several years to build a new rail link. By the time it is built, demand for rail transport may have changed. Transport infrastructure development may not always be well linked. New train stations may not be situated close to airports, for example. Infrastructure development may also have harmful effects on the environment.

The overall impact of technological improvement is often beneficial. However, its benefits are not always evenly spread and can be harmful to some. Technological improvement involves change. New products and new methods of production are created. New jobs appear while others are lost. Some people may have to change jobs and move to different areas, and not everyone will cope well with this.

Any supply-side policy tool which increases aggregate supply may not raise output if the economy is initially operating with spare capacity. This is because while it will increase productive potential, that potential will not be used if there is not enough aggregate demand. Firms may be capable of producing more, but they will not do so if they do not expect to sell any extra output.

TIP: Remember when answering questions that the impact of supply-side policy will be influenced by whether there is initially spare capacity in the economy.